

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1710

COURSE OUTCOME COVERED

CO3: Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)

Assessment Tool Weightage: 50%

Time Duration: 1 Hour

QUESTIONS: 4

Total Marks:40

Annexure A

CASE STUDY

Code of Conduct:

192565026
ShaiK Fahed (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

CASE STUDY

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Question 1

Case Study - Smart Medical Diagnosis System

A hospital is designing a rule-based Logical Agent for preliminary patient diagnosis. The system's knowledge base contains:

- I. If a patient has Fever AND Cough $\rightarrow$ Possible Flu
- II. If a patient has Fever AND Rash $\rightarrow$ Possible Measles
- III. If a patient has Cough AND Breathlessness $\rightarrow$ Possible Pneumonia
- IV. Facts about Patient A: Fever = True, Cough = True, Breathlessness = True

- (a) Represent the above rules and facts precisely using Propositional Logic symbols. Define each symbol used.
- (b) Apply Forward Chaining on the knowledge base to derive all possible diagnoses for Patient A. Show each inference step with the rule fired and fact derived.
- (c) The goal is to confirm 'Patient A has Pneumonia.' Apply Backward Chaining and draw the goal reduction tree to verify this goal.

Question 2

Case Study - Autonomous Traffic Management Agent

A smart city traffic controller uses a FOL-based knowledge base to manage emergency vehicle routing:

- I. for-all x : Vehicle(x) AND EmergencyType(x) $\rightarrow$ ClearPath(x) [Rule 1]
- II. for-all x : ClearPath(x) $\rightarrow$ GreenSignal(x) AND NOT RedSignal(x) [Rule 2]
- III. for-all x for-all y : Vehicle(x) AND Behind(x, y) AND GreenSignal(y) $\rightarrow$ Proceed(x) [Rule 3]
- IV. Facts: Vehicle(Ambulance), EmergencyType(Ambulance), Vehicle(CarA), Behind(CarA, Ambulance)

- (a) Apply Unification to match Rule 1 with the given facts. Show the complete substitution set θ (MGU) at each step.
- (b) Using the Resolution method, prove that 'Proceed(CarA)' is true. Convert relevant clauses to CNF and show each resolution step.
- (c) Analyze whether the current FOL knowledge base is sufficient to handle two simultaneous emergency vehicles. Identify what additional rules or facts are needed and justify logically.

Question 3

Case Study - Agricultural Expert Reasoning System

An agricultural advisory logical agent operates with the following propositional knowledge base (KB):

- 1) P1: SoilDry $\rightarrow$ IrrigationNeeded
- 2) P2: IrrigationNeeded AND CropWheat $\rightarrow$ ApplyDripMethod
- 3) P3: NOT ApplyDripMethod AND CropWheat $\rightarrow$ CropAtRisk
- 4) Facts: SoilDry = True, CropWheat = True

- (a) Convert all sentences P1, P2, P3 and the facts into Conjunctive Normal Form (CNF). Show each conversion step (eliminate implication $\rightarrow$ move NOT inward $\rightarrow$ distribute).
- (b) Using Resolution Refutation, prove the theorem 'ApplyDripMethod'. Negate the goal, combine with the KB in CNF, and derive the empty clause. Draw the resolution proof tree.
- (c) If the fact 'SoilDry' is removed from the KB, analyze step-by-step how logical conclusions change. Does 'CropAtRisk' still follow? Apply appropriate reasoning patterns to justify.

Question 4

Case Study - Wumpus World Knowledge Agent

An AI agent navigating the Wumpus World perceives: Stench at [1,2], Breeze at [1,1], Glitter at [2,2]. The agent uses a propositional KB with these rules:

- 1) Stench[1,2] $\rightarrow$ Wumpus is adjacent to cell [1,2]
- 2) Breeze[1,1] $\rightarrow$ Pit is adjacent to cell [1,1]
- 3) Glitter[x,y] $\rightarrow$ Gold is at [x,y]
- 4) NOT Wumpus[x,y] AND NOT Pit[x,y] $\rightarrow$ Safe[x,y]

- (a) Construct the agent's belief state from its percepts. Apply Modus Ponens systematically to derive new knowledge facts. List all derived conclusions.
- (b) Using Forward Chaining from the agent's percepts, identify the possible locations of the Wumpus and Pit. Show each reasoning step precisely.
- (c) Evaluate whether the path [1,1] $\rightarrow$ [1,2] $\rightarrow$ [2,2] is safe for the agent to collect the gold. Use the knowledge derived in (a) and (b) to justify your conclusion with logical inference.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks (100)
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly	24
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	23
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	19
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	19
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	9

(94)

SP

1. SMART Medical Diagnosis System

(a) propositional logic Representation

let

F = patient has Fever

C = Patient has cough

R = patient has Rash

B = patient has Breathlessness

Flu = Possible Flu

Measles = possible Measles

Pneu = possible pneumonia.

Rules :-

1) Fever AND Cough $\rightarrow$ possible Flu

$F \wedge C \rightarrow Flu$

2) Fever AND Rash $\rightarrow$ possible measles

$F \wedge R \rightarrow Measles$

3) Cough AND Breathlessness $\rightarrow$ possible pneumonia

$C \wedge B \rightarrow Pneu$

(b) Forward chaining:-

Initial Facts

① Fever, cough, Breathlessness

Fever $\wedge$ cough $\rightarrow$ Flu

Flu = True.

② cough $\wedge$ Breathlessness $\rightarrow$ Pneumonia

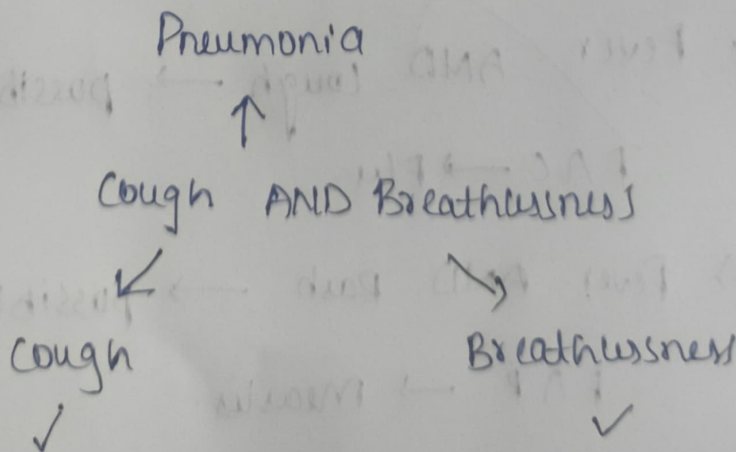
Pneumonia = True.

③ Fever $\wedge$ Rash $\rightarrow$ Measles

Rash is not given, so this rule can't fire

(c) Backward chaining of pneumonia:-

cough $\wedge$ Breathlessness $\rightarrow$ pneumonia



Autonomous Traffic Management Agent:-

(a) Unification with Rule 1.

Rule 1:-

Vehicle (x) $\wedge$ Emergency Type (x) $\longrightarrow$ clear path (x)

Facts:-

Vehicle (Ambulance)

Emergency Type (Ambulance)

Vehicle (carA)

Behind (carA, Ambulance)

$\therefore x = \text{Ambulance}$.

Emergency Type (Ambulance)

The MGU is

$\{x \mid \text{Ambulance}\}$

Apply

clear path (Ambulance)

So this rule fires.

(b) Resolution method for proceed (can't)

Relevant Rules:-

RS:-

clearpath (x) $\wedge$ GreenSignal (x) $\wedge$ $\neg$ RedSignal(x)
 $\rightarrow$ proceed (x)

CNF :-

$\neg$ clearpath(x) $\vee$ $\neg$ GreenSignal (x) $\vee$ RedSignal(x)
 $\vee$ proceed (x).

(c) Handling Two Simultaneous Emergency Vehicles:-

1) Emergency priority rule:-

Emergency Type(x) $\rightarrow$ priority (x)

2) Green signal (Ambulance 1)

Green signal (Ambulance 2)

3) proceed (x) $\wedge$ proceed (y) $\wedge$

Conflict (x,y) $\rightarrow$ stop (y)

3. Agricultural Expert Reasoning System

Given:-

- 1) Soil Dry $\rightarrow$ Irrigation Needed
- 2) Irrigation Needed $\wedge$ Crop wheat $\rightarrow$
Apply Drip method.
- 3) $\neg$ Apply Drip method $\wedge$ Crop wheat
 $\rightarrow$ Crop At Risk.

Facts:-

Soil Dry = True

Crop wheat = True

P₁ :- $\neg$ Soil Dry $\vee$ Irrigation Needed

P₂ :- $\neg$ Irrigation Needed $\vee \neg$ Crop wheat $\vee$
Apply Drip Method.

P₃ :- Apply Drip method $\vee \neg$ Crop wheat $\vee$
Crop at Risk

(b) Resolution Refutation to prove Apply Drip method

Goal :-

Negate the Goal :-

$\neg$ Apply Drip method

Clause :-

$\neg$ Irrigation Needed

$\vee$ $\neg$ Crop wheat $\vee$ Apply Drip

Result :-

$\neg$ Irrigation Needed $\vee$ $\neg$ Crop wheat

use fact & P. 1 :-

~~$\neg$ Soil Dry $\vee$ Irrigation Needed~~

Resolve with :-

$\neg$ Irrigation Needed

Result :-

$\neg$ Soil Dry

Apply Drip method

is approved.

Wumpus World Knowledge Agent:-

Given perceptions:-

Stench at $[1,2]$

Breeze at $[1,1]$

Glitter at $[2,2]$

Stench $[x,y] \rightarrow$ Wumpus is adjacent to $[x,y]$

Breeze $[x,y] \rightarrow$ pit is adjacent to $[x,y]$

Glitter $[x,y] \rightarrow$ Gold is at $[x,y]$

$\neg$ Wumpus $[x,y] \rightarrow$ Safe $[x,y]$.

(a) Belief state and modus ponens

Perception 1:-

Since stench $[1,2]$ is true:-

wumpus is adjacent to $[1,2]$

possible adjacent cells:-

$[1,1]$, $[1,3]$, $[2,2]$

Perception 2:-

Pit is adjacent to $[1,1]$

possible cells:- $[1,2]$, $[2,1]$

Perception 3: Glitter $[2,2]$

Glitter $[2,2] \rightarrow$ Gold $[2,2]$

Gold is at $[2,2]$

(b) Possible wumpus and pit locations

① Wumpus :- From Stench $[1,2]$

possible locations are:-

$[1,1]$, $[1,3]$, $[2,2]$

So

wumpus $\in \{[1,1], [1,3], [2,2]\}$

② pit :- From Breeze $[1,1]$

possible locations :-

$[1,2]$, $[2,1]$

So

pit $\in \{[1,2], [2,1]\}$

③

Gold :-

From :-

Glitter $[2,2]$

Therefore

Gold = $[2,2]$